

COMPUTER ARCHITECTURE - CO1 – ASSESSMENT TOOL – 2

Question 1

Concept Map

Video Editing Application



Input Unit (SSD)

↓ transfers data via

System Bus



Memory (RAM)



CPU

└─ Control Unit → Fetch → Decode

└─ ALU → Video Filters, Compression, Audio Processing



Output Unit

└─ Display Preview

└─ Storage Device

Explanation:

- Input unit reads multimedia data.
- Bus transfers instructions and data.
- RAM stores frames and instructions.
- CPU processes video/audio.
- Output unit displays and stores rendered video.

Question 2

Concept Map

Program



Fetch → Decode → Execute



Clock Cycles



CPI

↓
Clock Rate
↓
CPU Execution Time

Formula:

$$\text{CPU Execution Time} = (\text{Instruction Count} \times \text{CPI}) / \text{Clock Rate}$$

Explanation:

Fetch, Decode and Execute determine clock cycles. Lower CPI and higher clock rate improve performance.

Question 3

Concept Map

Microprocessor

- └─8085
 - | └8-bit
 - | └64 KB Memory
 - | └No Segmentation
 - | └Lower Speed
- └─8086
 - └16-bit
 - └1 MB Memory
 - └Memory Segmentation
 - └Higher Speed

Influence:

8086 offers faster execution, better memory management and multitasking.

Question 4

Concept Map

Addressing Modes

- └Immediate
- └Direct
- └Register
- └Indirect
 - └Indexed

↓
Data Transfer | Arithmetic | Control Instructions

↓

Execution Speed, Memory Access, Instruction Size

Applications:

Immediate-constants, Direct-fixed memory, Register-fast arithmetic,
Indirect-pointers, Indexed-arrays.

Question 5

Concept Map

Binary ↔ Hexadecimal

↓

Instruction Representation

↓

Memory Storage

↓

CPU Fetch → Decode → Execute

↓

Debugging

Example:

$$10101111_2 = AF_{16}$$

Explanation:

Binary is used internally while hexadecimal simplifies memory addresses and debugging.